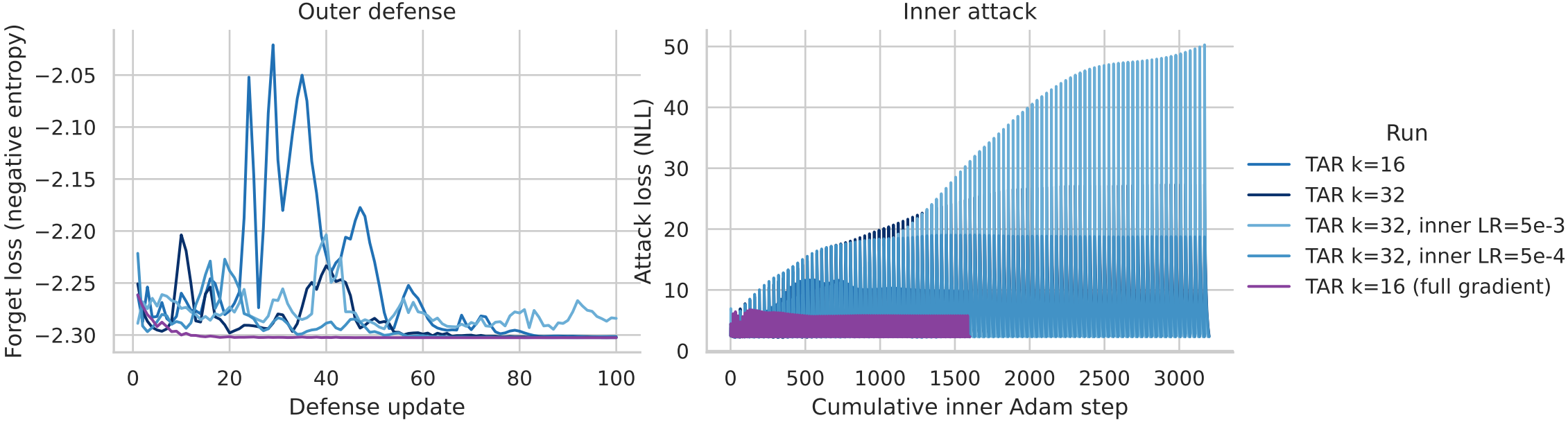

Outer defense
Forget loss (negative entropy)
−2.05
−2.10
−2.15
−2.20
−2.25
−2.30
0
20
40
60
80
100
Defense update
Inner attack
Attack loss (NLL)
50
40
30
20
10
0
0
500
1000
1500
2000
2500
3000
Cumulative inner Adam step
Run
TAR k=16
TAR k=32
TAR k=32, inner LR=5e-3
TAR k=32, inner LR=5e-4
TAR k=16 (full gradient)